Out-of-Distribution Detection in Computer Vision

Comparative benchmark with a simplified STOOD-X statistical approach

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Problem and Objective

Problem. A classifier trained on CIFAR-10 can still assign high confidence to images outside its training distribution.

Objective. Detect whether an input image is In-Distribution (ID) or Out-of-Distribution (OOD), while keeping the trained classifier fixed.

Project focus.

- Train ResNet-18 on CIFAR-10.
- Benchmark post-hoc OOD detectors.
- Partially replicate the feature-space intuition of STOOD-X.

CIFAR-10 is the known distribution.

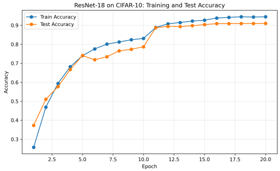
Experimental Setup

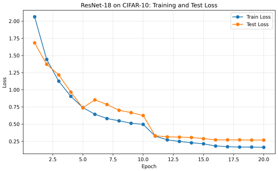
Datasets

- ID: CIFAR-10.
- OOD: CIFAR-100, SVHN, MNIST.
- All inputs normalized with CIFAR-10 statistics.

Classifier

- ResNet-18 adapted for 32×32 images.
- 20 training epochs.
- Best CIFAR-10 test accuracy: **91.03%**.





Metrics

- AUROC, AUPR-IN, AUPR-OUT: higher is better.
- FPR95: lower is better.

Methods Compared

Logit/confidence based

- **MSP**: maximum softmax probability.
- **Energy**: log-sum-exp score over logits.
- **ODIN**: temperature scaling plus input perturbation.

These methods use the classifier output directly.

Feature-distance based

- **Mahalanobis**: distance to class feature means.
- **KNN**: distance to the ID feature bank, $k = 50$.
- **STOOD-X simplified**: empirical p-value from KNN distances.

These methods use penultimate-layer representations.

Common convention: high score = more likely ID; low score = more likely OOD.

Simplified STOOD-X Procedure

1. Extract ResNet-18 features from CIFAR-10 training images.
2. Build an ID feature bank.
3. For each ID training sample, compute the distance to its k-th nearest neighbor.
4. Use these distances as the reference ID distribution.
5. For a new sample, convert its KNN distance into an empirical p-value.

$$p(x) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left(d_i^{ref} \geq d(x)\right)$$

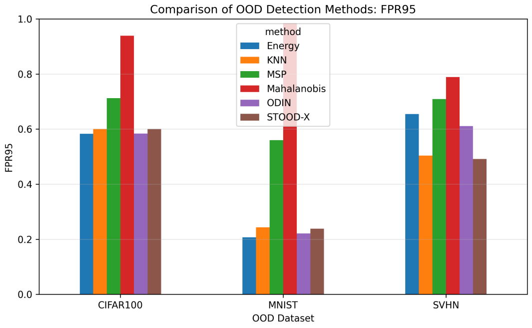
Interpretation

- High p-value: close to ID feature geometry.
- Low p-value: statistically unusual feature distance.

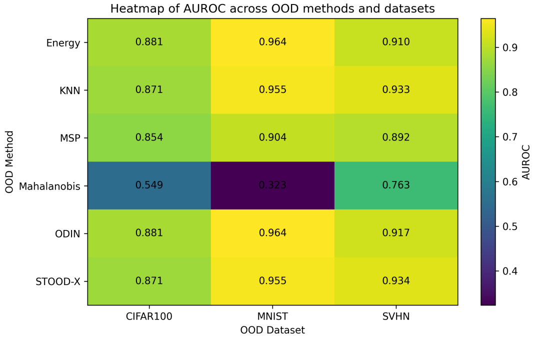
This is a partial replication: the full STOOD-X explainability stage and the original Wilcoxon-Mann-Whitney test were not implemented.

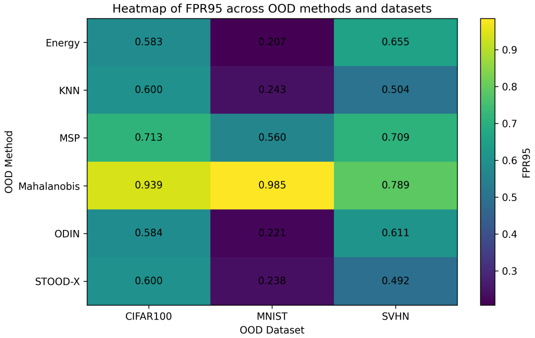
Quantitative Benchmark

OOD Dataset	Best AUROC	Best FPR95	Strongest Pattern
CIFAR-100	ODIN: **0.8810**	Energy: **0.5825**	hardest near-OOD case
MNIST	ODIN: **0.9643**	Energy: **0.2073**	easiest far-OOD case
SVHN	STOOD-X: **0.9339**	STOOD-X: **0.4916**	feature distances win



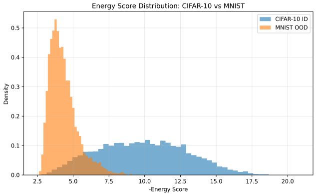
Full Result Matrix

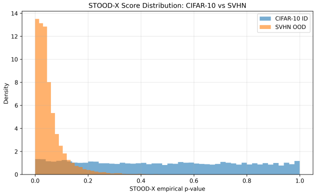


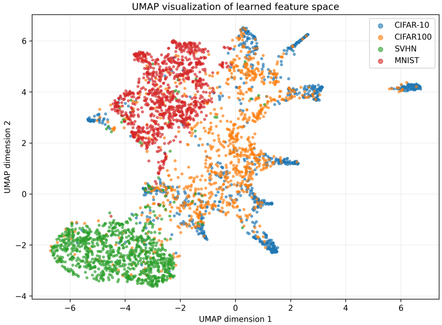


- ODIN and Energy are strongest on CIFAR-100 and MNIST.
- KNN and simplified STOOD-X are very close, because both rely on feature-space distances.
- Mahalanobis underperforms, especially on MNIST and CIFAR-100.

Score and Feature-Space Evidence







- Good detectors separate ID and OOD score distributions.
- UMAP confirms that feature geometry contains useful OOD information.

Main Conclusions

- The ResNet-18 classifier provides a strong feature space for post-hoc OOD detection.
- ODIN and Energy are the best overall logit-based methods.
- Simplified STOOD-X is strongest on SVHN and competitive on MNIST.
- Feature-distance methods are useful when OOD samples are not simply low-confidence inputs.
- The project produces reproducible outputs: metrics tables, ROC curves, score distributions, heatmaps, t-SNE/UMAP and a final report.

Limitations. Single architecture, single training run, three OOD datasets, and only a partial STOOD-X replication.

Takeaway: OOD detection improves when classifier confidence is combined with feature-space evidence.